

Answer Key — Phases of the Moon

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 (b) **reflects sunlight** — the Moon reflects sunlight; it has no light of its own.

Q2 (b) **waxing** — the bright part grows during waxing (Shukla Paksha).

Q3 (b) **new Moon** — the invisible Moon is the new Moon (Amavasya).

Q4 (b) **gibbous** — more than half lit is the gibbous phase.

Q5 (a) **True** — the cycle is about 29.5 days — nearly a month.

Q6 **purnima** — the full Moon day is Purnima.

Q7 **krishna** — the waning period is Krishna Paksha.

Short answer

Q8 The changing shapes of the Moon's bright portion seen from Earth; they occur because we see different amounts of its sunlit half as it revolves around Earth.

Q9 Waxing (Shukla Paksha): bright part grows from new to full Moon; waning (Krishna Paksha): bright part shrinks from full to new Moon.

Q10 (i) Full Moon (Purnima); (ii) crescent; (iii) New Moon (Amavasya).

Long / application

Q11 From New Moon (Amavasya) the bright part grows — crescent → half → gibbous → Full Moon (Purnima); this ~2-week growing period is Shukla Paksha (waxing). Then it shrinks — gibbous → half → crescent → New Moon; this ~2-week shrinking period is Krishna Paksha (waning). The whole cycle takes about 29.5 days.

Q12 The Moon's actual shape (a sphere) does not change. What changes is how much of its sunlit half we can see from Earth as it revolves, which makes the bright portion appear to change shape — the phases.
